Ex:13

8/10/26

Implement your own Ping Program

AIM:

To implement a simple ping program using python to check whether a destination host is reachable and to measure the round-trip time (RTT).

ALGORITHM:

* Start the program
* Get the destination hostname or IP address from the user.
* Create a raw ICMP socket.
* Construct an ICMP Echo Request packet and calculate its checksum.
* Send the packet to the destination host.
* Wait for the ICMP Echo Reply.
* Repeat the process for the four packets
* Display the output and terminate the program.

PROGRAM:

```
import socket
import struct
import time
import os
def checksums(data):
    total = 0
    for i in range(0, len(data)-1, 2):
```

```
    if len(data) % 2:
        total += data[-1] << 8
    total = (total >> 16) + (total & 0xffff)
    total += total >> 16
    return (~total) & 0xffff

def ping(host):
    try:
        dest_addr = socket.gethostbyname(host)
        icmp = socket.getprotobyname("icmp")
        sock = socket.socket(
            socket.AF_INET, socket.SOCK_RAW, icmp
        )
        sock.settimeout(2)
        packet_id = os.getpid() & 0xffff
        sequence = 1
        data = b"Computer Networks Ping"
        header = struct.pack(
            "!BBHHH", 8, 0, 0, packet_id, sequence
        )
        packet = header + data
        chksum = checksum(packet)
        header = struct.pack(
            "!BBHHH", 8, 0, chksum, packet_id, sequence
        )
        packet = header + data
        start = time.time()
        sock.recvfrom(packet, (dest_addr, 1))
        sock.recvfrom(1024)
        rtt = (time.time() - start) * 1000
        print("Reply from", dest_addr,
              ": bytes =", len(packet),
              "time = %.2f ms" % rtt)

        sock.close()
    except socket.timeout:
```

OUTPUT:

Enter hostname or IP address : 8.8.8.8

Reply from 8.8.8.8 : bytes = 30 time = 1.00ms

Reply from 8.8.8.8 : bytes = 30 time = 0.47ms

Reply from 8.8.8.8 : bytes = 30 time = 0.67ms

Reply from 8.8.8.8 : bytes = 30 time = 0.51ms

```
        print("Request timed out")
except PermissionError:
    print("Run with administrator/root privileges")
except socket.gaierror:
    print("Unable to resolve host")
host = input("Enter hostname or IP address: ")
for i in range(4):
    ping(host)
    time.sleep(1)
```

RESULT:

Thus, the ping program was successfully implemented using python to check network connectivity and calculate the round-trip time (RTT) of packets.